

CDetector: Extracting Textual Features of Financial Social Media to Detect Cyber Attacks

Jinwei Liu*, Long Cheng[†], Hongmei Chi*, Cong Liu[‡], Richard A. Alo*

*Department of Computer and Information Sciences, Florida A&M University, Tallahassee, FL 32307, USA

[†]School of Control and Computer Engineering, North China Electric Power University, Beijing, China

[‡]School of Computer Science and Technology, Shandong University of Technology, Shandong, China

*{jinwei.liu, hongmei.chi, richard.alo}@fam.u.edu, [†]lcheng@ncepu.edu.cn, [‡]liucongchina@sdu.edu.cn

Abstract—With the proliferation of social media, cyber threats and attacks have significantly increased in complexity and quantity in financial market. Malicious hackers leverage the influence of social media to spread deceptive information with an intent to gain abnormal profits illegally or to cause losses. Measuring information content in financial social media helps identify these threats and attacks. In this paper, we propose CDetector, an ML-based approach to identifying social media features that correlate with abnormal returns of the stocks of companies vulnerable to be targets of cyber attacks (e.g., cognitive hacking). To test our approach, we collected price data and the social media messages on multiple technology companies, and extracted features that contributed to abnormal stock movements. Preliminary results show that the top social media features associated with abnormal price movements are the terms that are simple, motivate actions, incite emotion, and use exaggeration, and the selected features correlate with abnormal messages and abnormal returns of the stocks of companies.

Index Terms—cyber attacks, cyber threats, abnormal behavior, cybersecurity, social media, feature extraction

I. INTRODUCTION

The prevalent use of social media facilitates the spreading of malicious messages that may inject misleading information to divert normal market operations [1]. Malicious hackers are increasingly exploiting the information environment and economic infrastructure to deceptively divert public sentiment to their own favors. The financial market thus becomes highly vulnerable to cybersecurity risks. Figure 1 shows examples of cyber attacks, and it reveals that misinformation (e.g., fake tweets) impacts the stock market. Cyber threats/attacks are still one of today's most challenging cybersecurity issues that are not well addressed by the commonly employed security solutions due to the rarity of confirmed positive threat cases and severe imbalance of ordinary datasets.

Machine learning attracts many interests and many studies utilize machine learning to identify cyber threats and attacks. Feature selection, as the process of selecting important features that contribute most to the learning model, hugely impacts the performance of the learning model. Efficient feature selection can significantly improve the accuracy and applicability of the learning and the process of classification [2]. In the data perspective, feature selection can be roughly categorized into four main categories: similarity-based methods, information-theoretical-based methods, statistical-based methods and sparse learning-based methods [3]. However,

the feature selection methods in these categories have some limitations [3]. To overcome the limitations of existing feature selection methods, we propose the hybrid feature selection (HFS). Financial social media data involves highly imbalanced occurrences and the skewed distribution datasets significantly impact the performance of the learning model [4]. In this paper, we develop CDetector to identifying social media features that correlate with abnormal returns of the stocks of companies vulnerable to be targets of cyber attacks. CDetector first uses the resampling method to balance the dataset, then it uses HFS to identify the social media features that correlate with abnormal returns of the stocks. Finally, it uses the machine learning algorithm to perform binary classification for detecting cyber attacks by utilizing the identified features.

The remainder of this paper is organized as follows. Section II reviews the related work. Section III presents the problem formulation and the description of data collection in this paper. Section IV describes the proposed ML-based approach to identifying social media features that correlate with abnormal returns of the stocks of companies vulnerable to be targets of cyber attacks. Section V presents the experimental results, findings and analyses. Section VI concludes this paper with remarks on our future work.

II. RELATED WORK

Cyber attack/threat detection has become an active research topic, and feature selection can possibly help to detect cyber threats/attacks. Many studies leverage feature selection to detect cyber threats/attacks. Below we review prior research on cyber attack/threat detection using feature selection.

Okutan *et al.* [5] used unconventional signals extracted from the Twitter and GDELT data sources and showed that they can be used to predict various cyber attacks for the anonymized target entity KNOX. They applied feature technique to the dataset with 10 x 10 folds cross validation to cross check their findings about the importance of each unconventional signal for each attack type. To balance the distribution of the majority and minority instances in an imbalanced dataset, they developed SMOTE++ that uses a hybrid approach where under sampling, synthetic minority instance generation and reweighing techniques are used together. Bilge *et al.* [4] proposed a set of unique features (i.e., network, activity, user, and tweet content) based on tweet contents and tweet information, and

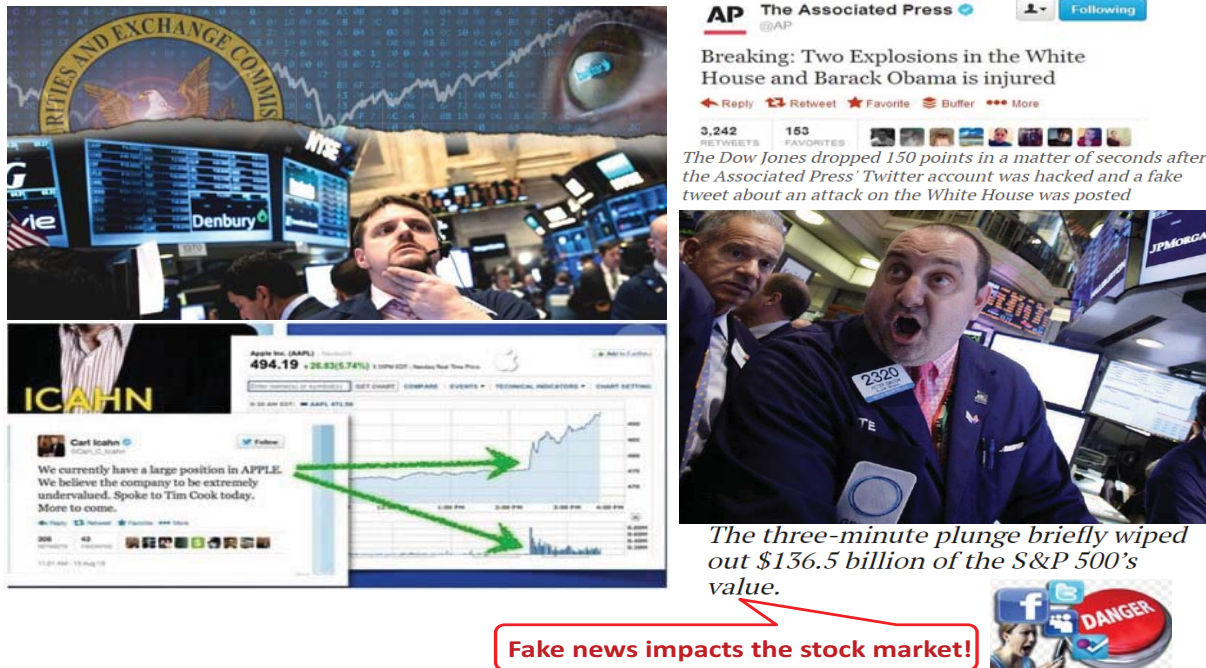


Fig. 1: Examples of fake tweets: fake news impacts stock market and can be dangerous.

they developed a supervised machine learning solution for detecting cyberbullying in the Twitter. They employed three feature selection algorithms (i.e., chi-square test, information gain and Pearson correlation) to determine the discriminative power of each feature, and they identified top 10 significant features. Hernandez-Suarez *et al.* [6] proposed a methodology for predicting cyber-attacks based on scraping and classifying Twitter data. The methodology collects historical tweets and classifies them as negative, positive and security-oriented, and it predicts cyber-attacks by using ℓ_1 regularization. Sabottke *et al.* [7] conducted a quantitative and qualitative exploration of the vulnerability-related information disseminated on Twitter. They used the mutual information for feature selection, and identified features that are useful for detecting exploits. Also, they designed and evaluated a detector for real-world exploits utilizing features extracted from Twitter data. Yao *et al.* [8] formulated cyberbullying detection as a sequential hypothesis testing problem, and leveraged the feature selection to design an algorithm for reducing the time to raise a cyberbullying alert by minimizing the number of feature evaluations necessary for a decision to be made.

However, most proportion of the prior feature selection methods is usually based on one chosen approach only without considering combining dissimilar feature selection approaches to enhance the performance. Also, many of the approaches neglect the nature of the financial social media data: highly imbalanced dataset. In this paper, we first use the resampling techniques to balance the dataset and clean the dataset. Then we leverage multiple dissimilar feature selection approaches to complement the errors made by singular mechanism, and propose a hybrid feature selection approach to extract texture features for detecting cyber attacks in financial social media.

III. PROBLEM STATEMENT AND DATA DESCRIPTION

In this section, we first formulate our problem, then we describe the data collection in this research. Table I shows the main notations used in the paper.

A. Problem Statement

Given the training data from social media messages and stock prices of technology companies (e.g., Apple, Microsoft, Intel, Cisco, IBM and VZ), how to effectively identify social media features that correlate with abnormal returns of the stocks of companies vulnerable to be targets of cyber attacks?

B. Data Collection

We develop the data collection by downloading the social media messages and stock prices of four technology companies: Apple, Microsoft, Intel and Cisco from July 2017 to August 2018 and the social media messages and stock prices of six technology companies: Apple, Microsoft, Intel, Cisco, IBM and VZ from January 2019 to June 2019. The stock prices and social media messages were crawled once every five minutes on each US trading day. The messages were collected from the sites Twitter and StockTwits by using their public APIs. Also, we collected numerical financial data of the aforementioned six companies from the following three sources: Google Finance, US Treasury and Yahoo Finance.

IV. IDENTIFYING SOCIAL MEDIA FEATURES

To detect cyber attacks in financial social media, we develop CDetector: an ML-based approach to identifying social media features that correlate with abnormal returns of the stocks of companies vulnerable to be targets of cyber attacks. In this section, we first describe the calculation of abnormal returns, then we describe the approach we developed in this research.

TABLE I: Notations

P_t	weighted portfolio's price at time t	R_f	Risk-free rate of return normalized to the time span of the scenario
β	portfolio i 's price volatility relative to the overall market	R_m	market return based on S&P 500 index normalized to the time span of the scenario
R_i	portfolio i 's actual rate of return $\frac{P_t - P_{t-1}}{P_{t-1}}$	$tf(t, d)$	term frequency
f_i	feature i	$IG(f_i)$	information gain of f_i
$idf(t, D)$	inverse document frequency	N	Total # of documents
$ \{d \in D : t \in d\} $	# of documents where term t appears	E_I	value of the entropy of the data
I_v	# of items in which f_i has a value equaling v	E_{I_v}	entropy computed on data where f_i assumes value v

A. Calculation of Abnormal Returns

We computed the abnormal rate of return in the context of a market of S&P 500 stocks. The returns of these stocks were weighed by their market capitalizations. We generated a label for each five-minute scenario by computing the abnormal return for the stock portfolio i as $R_i - [R_f + \beta \cdot (R_m - R_f)]$ where R_i is the portfolio i 's actual rate of return ($\frac{P_t - P_{t-1}}{P_{t-1}}$), and P_t is the weighted portfolio's price at time t ; R_f is the risk-free rate of return normalized to the time span of the scenario; R_m is the market return based on S&P 500 index normalized to the time span of the scenario; β is the portfolio i 's price volatility (over the past 36 months) relative to the overall market, and it is computed as the covariance of (Rate of Return of Stock, Rate of Return of Market) divided by market variance (where market is proxied by the S&P 500 index). Each scenario contains social media messages published during a five-minute window on a financial trading day (from 9:30 am to 4:00 pm). For each scenario, we assigned a label of 1 to the scenario if the absolute value of the abnormal rate of return is 0.5% or above; otherwise, we assigned a label of 0 to the scenario.

B. Extracting Social Media Features

We focus on the scenarios with abnormal return for identifying the associated social media features. We first remove URLs, punctuations and special characters, and then tokenize the messages, convert the messages to lowercase, and perform stemming.

Feature Identification using TFIDF: TFIDF is computed using Eqs. (1)-(3) below:

$$tf(t, d) = 0.5 + 0.5 \cdot \frac{f_{t,d}}{\max\{f_{t',d} : t' \in d\}} \quad (1)$$

$$idf(t, D) = \log \frac{N}{|\{d \in D : t \in d\}|} \quad (2)$$

$$tfidf(t, d, D) = tf(t, d) \cdot idf(t, D) \quad (3)$$

where $tf(t, d)$ is term frequency which measures the count of terms in a document, and $idf(t, D)$ is the inverse document frequency which measures how much information the word provides. In Formula (2), N is the total number of documents in the messages $N = |D|$, $|\{d \in D : t \in d\}|$ is the number of documents where the term t appears. If the term is not in the corpus, we adjust the denominator to $1 + |\{d \in D : t \in d\}|$ to avoid division-by-zero problem. In the experiment, we considered each message as a document, and all the messages

in the entire collection of scenarios that had abnormal price movements.

Feature Identification using Information Gain: We also use Information Gain (IG) to identify the social media features that correlate with abnormal returns of the stocks. IG is defined as the expected reduction in entropy occurring when a feature is present versus when it is absent. IG falls into the category of filter methods. Given a particular and arbitrary feature f_i , we compute the IG as follows [9]

$$IG(f_i) = E(I) - \sum_{v \in \text{dom}(f_i)} \frac{|I_v|}{|I|} \cdot E(I_v) \quad (4)$$

where $E(I)$ is the value of the entropy of the data, $|I_v|$ is the number of items in which the feature f_i , has a value equaling v , and $E(I_v)$ is entropy computed on data where the feature f_i assumes value v . In the experiment, we first computed the IG values of each term using Formula (4) and ranked the terms based on the IG value in descending order. Then we considered the terms with the top rank of IG values as the social media features that correlate with abnormal returns of the stocks.

Feature Identification using Gain Ratio: Gain ratio (GR) is an extension to information gain (IG) that reduces the bias of IG on highly branching features. GR considers the number and size of branches when it chooses an attribute. It corrects the IG by taking the intrinsic information of a split into account. Intrinsic information is entropy of distribution of instances into branches (i.e., how much information do we need to tell which branch an instance belongs to). As intrinsic information gets larger, the value of attribute decreases [10].

$$GR(\text{Attribute}) = \frac{\text{Gain}(\text{Attribute})}{\text{Intrinsic_info}(\text{Attribute})} \quad (5)$$

Feature Identification using Recursive Feature Elimination: Recursive feature elimination (RFE) is a feature selection method that fits a model and removes the weakest feature (or features) until the specified number of features is reached. RFE begins by building a model on the entire set of predictors and computing an importance score for each predictor. The least important predictor(s) are then removed, the model is re-built, and importance scores are computed again. RFE falls into the category of wrapper methods. The wrapper methods usually result in better predictive accuracy than filter methods. The feature selection in wrapper methods follows a greedy search approach by evaluating all the possible combinations of features against the evaluation criterion.

After calculating score of each feature using TFIDF, Information Gain, Gain Ratio and RFE, CDetector normalizes the

scores to the same range ([0,1]). Denote R_1^* as the normalized TFIDF score of a feature, R_2^* as the normalized IG score of the feature, R_3^* as the normalized GR score of the feature, and R_4^* as the normalized RFE score of the feature. Thus the overall rank of the feature can be obtained as follows:

$$R = \sum_{i=1}^m w_i R_i^* \quad (6)$$

where $m = 4$, and w_i ($i = 1, \dots, m$) are the weights.

To determine the weight of each factor (w_i), we use relative comparison method [11]. Let $H = (h_{ij})_{m \times m}$, $i, j = 1, 2, \dots, m$

$$h_{ij} = \begin{cases} 1, & \text{if } R_i^* \text{ is more important than } R_j^* \\ 0.5, & \text{if } R_i^* \text{ is the same important as } R_j^* \\ 0, & \text{if } R_i^* \text{ is less important than } R_j^* \end{cases} \quad (7)$$

Obviously:

$$\begin{cases} h_{ii} = 0.5 \\ h_{ij} + h_{ji} = 1 \end{cases} \quad (8)$$

Since the weight of each factor is related to the practical application, largely depending on the goal of the system, we can qualitatively assign weights to $R_1^*, R_2^*, R_3^*, R_4^*$ according to the goal of the system and determine the order of weight for each factor. The weight can be calculated by the following formula:

$$w_i = \frac{\sum_{j=1}^m h_{ij}}{\sum_{i=1}^m \sum_{j=1}^m h_{ij}} \quad (9)$$

In this way, we quantify the qualitative weight of each factor in Formula (6).

C. The Design of CDetector

In this section, we present the design of CDetector for detecting cyber attacks in financial social media. CDetector first identifies social media features by using HFS, then it uses the machine algorithm to perform classification using the identified features. The HFS is an important component of CDetector. Figure 2 shows the architecture of CDetector. CDetector consists of four phases: data preprocessing (Phase 1), feature selection (Phase 2), training and testing (Phase 3), determining cyber attacks (Phase 4). Below we describe each phase.

In Phase 1, CDetector employs the SMOTE algorithm [12] and the T-Link algorithm [13] to balance the dataset and clean the data. In Phase 2, CDetector leverages HFS to extract social media features that correlate with abnormal returns of the stocks (see Section IV-B). In Phase 3, CDetector performs training based on the identified features, and uses cross-validation for testing. In Phase 4, CDetector uses the machine learning algorithm to detect cyber attacks.

Algorithm 1 shows the pseudocode for CDetector. The algorithm first uses the resampling techniques (i.e., SMOTE and T-Link) to balance the dataset and clean the data (line 2). Next, it generates the weights for different feature selection methods (line 3). After generating the weights for different feature selection methods, the algorithm uses HFS to identify top K features that correlate with abnormal returns of stocks (line 3). Next, the algorithm performs training based on the identified top K features (line 4). Then, the algorithm uses

Algorithm 1: Pseudocode for CDetector

Input: Social media messages (tweets) and stock prices of four technology companies and six technology companies

Output: The categories of social media tweets

- 1 Performs data cleaning (e.g., removing URLs, punctuations and special characters, and then tokenizing the messages, etc.)
- 2 Balance the dataset using the SMOTE algorithm and T-Link algorithm
- 3 Use HFS to identify top K features that correlate with abnormal returns of stocks // Section IV-B
- 4 Train the algorithm based on the identified top K features
- 5 Use cross-validation for testing
- 6 Predict if the tweets are cyber threats/attacks by predicting if the stock has abnormal return

cross-validation for testing (line 5). Finally, the algorithm uses the machine learning algorithm to predict if the tweets are cyber threats/attacks by predicting if the stock has abnormal return (line 6).

V. EXPERIMENTAL FINDINGS

In this section, we present the experimental findings of an empirical study of two datasets. Dataset 1 was collected from July 2017 to August 2018. The data consists of 10,301 scenarios (containing 865,289 messages). Then, we chose the scenarios with abnormal returns to analyze features that correlate with abnormal returns of the stocks, which consist of 537 scenarios (containing 14,843 messages). The proportion of the abnormal messages is 1.71%. Prices of the stocks at the beginning and end of scenarios were used to compute returns.

Dataset 2 was collected from January 2019 to June 2019. The data consists of 2,302 scenarios (containing 105,402 messages). Then, we chose the scenarios with abnormal returns to analyze features that correlate with abnormal returns of the stocks, which consist of 69 scenarios (containing 1,053 messages). The proportion of the abnormal messages is 1%.

We conducted experiments to identify the social media features that correlate with abnormal returns of the stocks to detect cyber attacks based on different methods.

For the implementation of TFIDF, we used the python libraries pandas, numpy, nltk for data processing, and we used library scikit-learn to extract features. In the implementation of IG, we considered two categories: positive and negative. We first calculated the probabilities of different variables (e.g., prob. of a category C_i , prob. of a term, prob. of absence of a term, etc.) to compute the IG of each term. Finally, we calculated the IG values for each term and sorted the values in descending order and terms with top rank of IG values were selected as the target social media features that correlate with abnormal returns of stocks. Table II shows the representative messages from normal and abnormal scenarios in Dataset 1 and Dataset 2.

Table III shows the top 20 features identified by each method based on Dataset 1. The last row shows features commonly identified by all three methods. We see that the identified features are emotion-oriented and typically reflect the status/trend of stocks. These messages are likely to encourage or discourage people to take actions and therefore have the

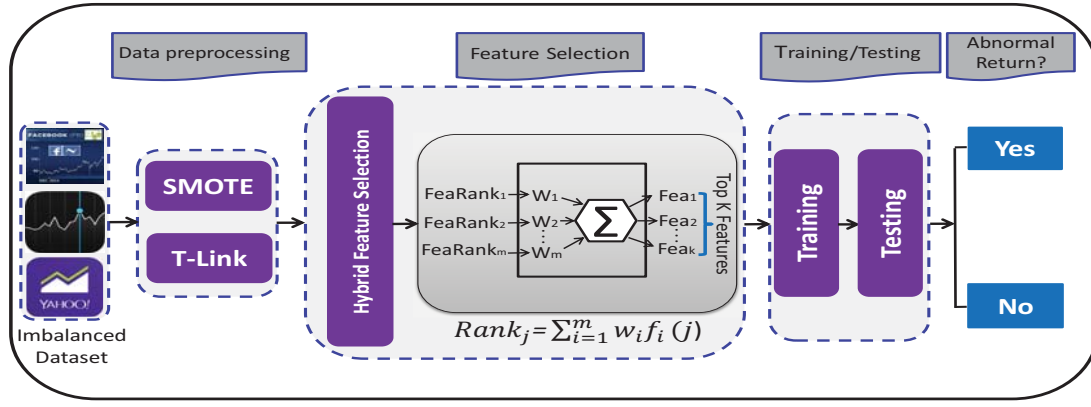


Fig. 2: The architecture of CDetector.

TABLE II: Representative messages with abnormal stock returns and normal stock returns.

Normal Messages From Dataset 1
M1: \$INTC how long is this borefest?
M2: \$AAPL I am just follow Buffet.
M3: \$MSFT about to pop that viagra pill.
M4: \$AAPL Watch this ramp to 149 in a matter of 24 hrs lol.
M5: \$INTC Hahaha no benchmarks...
M6: \$AMZN, \$TSLA, \$GOOG, \$NVDA, \$AAPL so? just another story of the day.
Abnormal Messages From Dataset 1
M1: No real selling allowed yet. Buy, buy, buy. \$AAPL \$AMZN AMZN
M2: \$AAPL You want to make a few bucks, buy now and sell around 170.50
M3: \$AAPL follow big money buffet and don't sell but look to buy dips & accumulate
M4: \$AAPL y'all better buy ASAP! This is going to 180 by Friday if not higher..
M5: \$AAPL buy the dip, soon rising
M6: \$AAPL Market is "crashing" hard." But keep calm and buy buy buy!
Normal Messages From Dataset 2
M1: \$AAPL thanks god
M2: \$AAPL \$AMZN like I said this morning....
M3: \$AAPL has released the second iOS 11.3 beta for public testers.
M4: \$AAPL normal service resumed chaps
M5: My shopping list today: \$AAPL \$MU \$TXT \$CY \$JNJ \$UNH \$URI \$CMI \$LYB \$DXJ \$EEM \$IWM \$SCSO \$ANET This is what happens when \$AMZN use their own switches...
M6: Time to load on some quality stocks \$AMZN \$GOOGL \$FB \$NFLX \$INTC
Abnormal Messages From Dataset 2
M1: \$AAPL so strong.. sell the house and buy AAPL and can sleep well in the car!
M2: \$AAPL omg. Sell your kidney, first born and buy buy buy.
M3: \$BABA sell and buy \$AAPL near/OTM calls and you'll have made all your money back in less than 10 days.
M4: \$AAPL cheap stock price is getting and getting more attractive to scoop up big! Then one of these days, it will gap up to 229+! Cheap Buy!!!
M5: \$INTC over sold on all metrics buy buy buy
M6: \$INTC Bogus press release from CEO on Friday starting to sink in longs? Extremely bearish! Sell your shares buy \$AMD and thank me later!

potential of incurring abnormal returns of stocks. Also, some features are identified by all three methods, and hence are more likely to be associated with abnormal stock price movement.

Table IV shows the top 20 features identified by each method based on Dataset 2. The last row shows features commonly identified by all three methods. Similarly, we see that the identified features are emotion-oriented and typically

TABLE III: Top 20 features correlate with abnormal returns of the stocks using different methods based on Dataset 1.

Method	Feature
TFIDF	Buy, Sell, Good, Like, High, Low, Price, Grow, Rise, Great, Earnings, Short, Back, Trace, Hedging, Dip, Coming, Top, Bullish, Strong
IG	Yet, Buy, Fall, Short, Price, Bear, Crap, Boom, Fear, Divid, Down, Great, Loose, Love, Miss, Broke, Cheap, Never, Increase, Rise
HFS	Buy, Price, Sell, Great, Rise, Short, Down, Increase, Low, Cheap, Bullish, Miss, Strong, Yet, Boom, Cut, CEO, High, Crap, Close
TFIDF $\cap$ IG $\cap$ HFS	Buy, Price, Rise, Great

TABLE IV: Top 20 features correlate with abnormal returns of the stocks using different methods based on Dataset 2.

Method	Feature
TFIDF	Like, Buy, Open, Free, Good, Weak, Earnings, Short, Nice, CEO, Bob, Great, Trade, Back, Active, Floor, Big, Trump, Coming, Low
IG	Bob, More, Free, Down, Close, Easte, But, Cut, Like, Open, Build, Floor, CEO, Good, Alert, Cleve, Low, Buy, Out, Great
HFS	Buy, Like, Open, Free, Good, Floor, Great, CEO, Close, Nice, Down, Earnings, Cut, Up, Back, Sell, Rise, Price, Out, Strong
TFIDF $\cap$ IG $\cap$ HFS	Like, Buy, Open, Free, Good, CEO, Great, Floor

reflect the status/trend of stocks, and the messages are likely to encourage or discourage people to take actions and therefore have the potential of incurring abnormal returns of stocks. Compared to the results in Table III, more common features are identified by all three methods. The features identified in these tables may be helpful to indicate risks of cyber attacks.

A. Relationship between Top Features and Abnormal Messages

To test if the identified features are correlated with abnormal returns of stocks, we also tested the correlation between the occurrence of top features identified by TFIDF and IG and the percentage of abnormal messages based on our two datasets. Figure 3 shows the correlation between the occurrence of top features and the percentage of abnormal messages in TFIDF and IG based on Dataset 1. In Figure 3, we see that the percentage of abnormal messages increases as the percentage of the occurrence of top features increases. For the same percentage of occurrence of top features, the percentage of abnormal

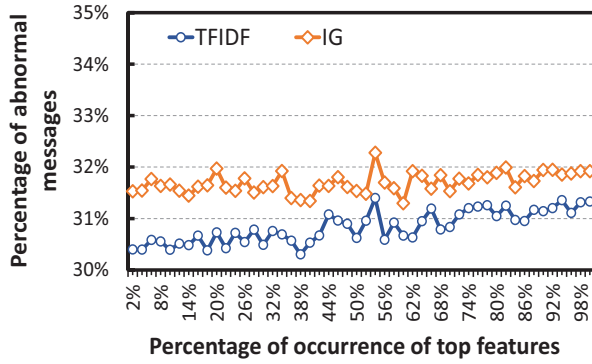


Fig. 3: Relationship between top features and abnormal messages based on Dataset 1.

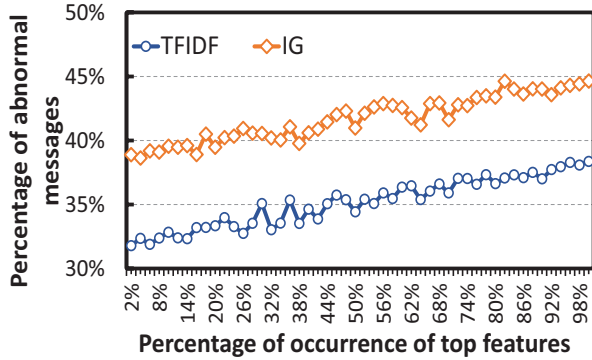


Fig. 4: Relationship between top features and abnormal messages based on Dataset 2.

message in IG is relatively higher than that of TFIDF. This demonstrates that the identified features are correlated with abnormal returns of stocks, and the performance of IG is relatively better than that of TFIDF. Figure 4 shows the correlation between the occurrence of top features and the percentage of abnormal messages in TFIDF and IG based on Dataset 2. In Figure 4, we see similar results. By examining Figure 3 and Figure 4, we find that, for the same percentage of occurrence of top features, the percentage of abnormal message in IG and TFIDF in Figure 4 is relatively higher than that in Figure 3. This indicates that the identified features based on Dataset 2 are more correlated with abnormal returns of stocks.

To test the significance of the correlation between the percentage of abnormal messages and the occurrence of top features, we conducted experiments based on Dataset 2 and tested the p-value of the correlation for the methods shown in Figure 4. Table V shows the significance test of the correlation between the percentage of abnormal messages and the percentage of the occurrence of top features in different methods. The experimental results show that there is a significant positive relationship between the percentage of abnormal messages and the percentage of the occurrence of top features, $r(48)_{TFIDF} = 0.938$, $r(48)_{IG} = 0.945$, $p_{TFIDF} < 0.001$, $p_{IG} < 0.001$.

VI. CONCLUSIONS

In this paper, we develop CDetector, an ML-based approach to identifying social media features that correlate with abnormal returns of the stocks of companies vulnerable to be targets

TABLE V: Significance test of the correlation between the percentage of abnormal messages and the percentage of the occurrence of top features in different methods.

Method	Pearson Correlation Coefficient	R Square	p-value (5% level)	Statistical Significance (5% level)
TFIDF	0.938	0.880	1.026E-23	Extremely Significant
IG	0.945	0.894	5.202E-25	Extremely Significant

of cyber threats and attacks. Preliminary results show that terms that are simple, motivate actions, incite emotion, and use exaggeration are ranked high in the features of messages associated with abnormal price movements. We also provide selected messages to illustrate the use of these features in potential cyber attacks. This research should contribute to a new approach to understanding specific features that may associate with cyber attacks and to providing new empirical findings of examining social media and their abnormal effect on financial market. In the future, we will expand the dataset and provide the experimental results of quantitative success metrics and give more solid evidence of the effectiveness of our approach (including HFS). We will also compare our approach with state-of-the-art to fully verify the performance of CDetector.

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